

Per-Element Trace Contributions for Diagonal K3 Surfaces

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For each of the 14 diagonal K3 surfaces (Goto Table 7), we list all orbifold cohomology basis elements with their twisted Frobenius eigenvalues α_i .

Surface #1: $Q = (21, 14, 6, 1)$, $m = 42$

Equation: $x_0^2 + x_1^3 + x_2^7 + x_3^{42} = 0$

$p = 3529$

Table 1: Trace contributions for $x_0^2 + x_1^3 + x_2^7 + x_3^{42} = 0$ at $p = 3529$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{42}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{42}\right)$	$1146+1267\cdot 3529+2875\cdot 3529^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^{37}$	p	p	3529
	$y_1^1 y_2^1 y_3^3 y_4^{31}$	p	p	3529
	$y_1^1 y_2^1 y_3^4 y_4^{25}$	p	p	3529
	$y_1^1 y_2^1 y_3^5 y_4^{19}$	p	p	3529
	$y_1^1 y_2^1 y_3^6 y_4^{13}$	p	p	3529
	$y_1^1 y_2^1 y_3^7 y_4^{11}$	p	p	3529
	$y_1^1 y_2^1 y_3^8 y_4^{17}$	p	p	3529
	$y_1^1 y_2^1 y_3^9 y_4^{13}$	p	p	3529
	$y_1^1 y_2^1 y_3^{10} y_4^{11}$	p	p	3529
	$y_1^1 y_2^1 y_3^{11} y_4^5$	p	p	3529
	$x_1^1 x_2^1 x_3^2 x_4^{37} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{37}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{37}{42}\right)$	$441\cdot 3529+3300\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^3 x_4^{31} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{31}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{31}{42}\right)$	$3236\cdot 3529+3414\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^4 x_4^{25} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{25}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{25}{42}\right)$	$3325\cdot 3529+3228\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^5 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{19}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{19}{42}\right)$	$276\cdot 3529+2589\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^6 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{13}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{13}{42}\right)$	$1761\cdot 3529+2946\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^7 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{7}\right)\Gamma_p\left(\frac{11}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{7}\right)\Gamma_p\left(\frac{11}{42}\right)$	$1008\cdot 3529+2640\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^8 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{7}\right)\Gamma_p\left(\frac{17}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{7}\right)\Gamma_p\left(\frac{17}{42}\right)$	$1675\cdot 3529+1651\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^9 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{9}{7}\right)\Gamma_p\left(\frac{13}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{9}{7}\right)\Gamma_p\left(\frac{13}{42}\right)$	$1678\cdot 3529+663\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^{10} x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{10}{7}\right)\Gamma_p\left(\frac{11}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{10}{7}\right)\Gamma_p\left(\frac{11}{42}\right)$	$2987\cdot 3529+2101\cdot 3529^2+\dots$
	$x_1^1 x_2^1 x_3^{11} x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{7}\right)\Gamma_p\left(\frac{5}{42}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{7}\right)\Gamma_p\left(\frac{5}{42}\right)$	$3521\cdot 3529+531\cdot 3529^2+\dots$
$H^{2,0}$	$x_1^1 x_2^2 x_3^8 x_4^{41} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{41}{42}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{41}{42}\right)$	$2288\cdot 3529^2+\dots$
$H^{2,2}$	$y_1^1 y_2^2 y_3^6 y_4^{41}$	p^2	p^2	12453841

Surface #2: $Q = (15, 10, 3, 2)$, $m = 30$

Equation: $x_0^2 + x_1^3 + x_2^{10} + x_3^{15} = 0$

$p = 1801$

Table 2: Trace contributions for $x_0^2 + x_1^3 + x_2^{10} + x_3^{15} = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{15}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{15}\right)$	$72+845\cdot 1801+1550\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^3 y_4^{13}$	p	p	1801
	$y_1^1 y_2^1 y_3^3 y_4^{10}$	p	p	1801
	$y_1^1 y_2^1 y_3^7 y_4^7$	p	p	1801
	$y_1^1 y_2^1 y_3^9 y_4^4$	p	p	1801
	$y_1^1 y_2^2 y_3^5 y_4^{11}$	p	p	1801
	$y_1^1 y_2^2 y_3^8 y_4^8$	p	p	1801
	$y_1^1 y_2^2 y_3^5 y_4^5$	p	p	1801
	$y_1^1 y_2^2 y_3^7 y_4^2$	p	p	1801
	$x_2^1 x_4^{10} y_1^1 y_3^5 e_2 e_4$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_2^1 x_4^5 y_1^1 y_3^5 e_2 e_4$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^1 x_3^5 y_2^1 y_4^{10} e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^1 x_3^5 y_2^2 y_4^5 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^1 x_2^1 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{13}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{13}{15}\right)$	$1416\cdot 1801+391\cdot 1801^2+\dots$
	$x_1^1 x_2^1 x_3^5 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^1 x_2^1 x_3^7 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{7}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{7}{15}\right)$	$1081\cdot 1801+438\cdot 1801^2+\dots$
	$x_1^1 x_2^2 x_3^4 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{11}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{11}{15}\right)$	$409\cdot 1801+1590\cdot 1801^2+\dots$
	$x_1^1 x_2^2 x_3^4 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{8}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{8}{15}\right)$	$1026\cdot 1801+1383\cdot 1801^2+\dots$
	$x_1^1 x_2^2 x_3^8 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{8}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{8}{15}\right)$	$903\cdot 1801+410\cdot 1801^2+\dots$
	$x_1^1 x_2^2 x_3^5 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^1 x_2^2 x_3^7 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{15}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{15}\right)$	$1525\cdot 1801+1463\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^1 x_2^2 x_3^9 x_4^{14} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{14}{15}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{14}{15}\right)$	$1776\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^1 y_2^2 y_3^9 y_4^{14}$	p^2	p^2	3243601

Surface #3: $Q = (12, 8, 3, 1)$, $m = 24$

Equation: $x_0^2 + x_1^3 + x_2^8 + x_3^{24} = 0$

$p = 1153$

Table 3: Trace contributions for $x_0^2 + x_1^3 + x_2^8 + x_3^{24} = 0$ at $p = 1153$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{24}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{24}\right)$	$606+166\cdot 1153+854\cdot 1153^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^3 y_4^{19}$	p	p	1153
	$y_1^1 y_2^1 y_3^3 y_4^{13}$	p	p	1153
	$y_1^1 y_2^1 y_3^7 y_4^7$	p	p	1153
	$y_1^1 y_2^2 y_3^1 y_4^{17}$	p	p	1153
	$y_1^1 y_2^2 y_3^3 y_4^{11}$	p	p	1153
	$y_1^1 y_2^5 y_3^5 y_4^5$	p	p	1153
	$x_1^1 x_3^4 y_2^1 y_4^{16} e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1153
	$x_1^1 x_3^4 y_2^2 y_4^8 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1153
	$x_1^1 x_2^1 x_3^2 x_4^{22} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{11}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{11}{24}\right)$	$140\cdot 1153+37\cdot 1153^2+\dots$
	$x_1^1 x_2^1 x_3^3 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{19}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{19}{24}\right)$	$141\cdot 1153+252\cdot 1153^2+\dots$
	$x_1^1 x_2^1 x_3^4 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{2}{3}\right)$	1153
	$x_1^1 x_2^1 x_3^5 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{13}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{13}{24}\right)$	$25\cdot 1153+696\cdot 1153^2+\dots$
	$x_1^1 x_2^1 x_3^6 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{4}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{4}\right)\Gamma_p\left(\frac{5}{12}\right)$	$1013\cdot 1153+1115\cdot 1153^2+\dots$
	$x_1^1 x_2^1 x_3^7 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{24}\right)$	$1062\cdot 1153+813\cdot 1153^2+\dots$
	$x_1^1 x_2^2 x_3^1 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{17}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{17}{24}\right)$	$38\cdot 1153+758\cdot 1153^2+\dots$
	$x_1^1 x_2^2 x_3^2 x_4^{14} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{12}\right)$	$140\cdot 1153+37\cdot 1153^2+\dots$
	$x_1^1 x_2^2 x_3^3 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{11}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{11}{24}\right)$	$369\cdot 1153+980\cdot 1153^2+\dots$
	$x_1^1 x_2^2 x_3^4 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	1153
	$x_1^1 x_2^2 x_3^5 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{24}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{24}\right)$	$507\cdot 1153+42\cdot 1153^2+\dots$
	$x_1^1 x_2^2 x_3^6 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$1013\cdot 1153+1115\cdot 1153^2+\dots$
$H^{2,0}$	$x_1^1 x_2^2 x_3^7 x_4^{23} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{23}{24}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{23}{24}\right)$	$215\cdot 1153^2+\dots$
$H^{2,2}$	$y_1^1 y_2^2 y_3^7 y_4^{23}$	p^2	p^2	1329409

Surface #4: $Q = (10, 5, 4, 1)$, $m = 20$

Equation: $x_0^2 + x_1^4 + x_2^5 + x_3^{20} = 0$

$p = 1601$

Table 4: Trace contributions for $x_0^2 + x_1^4 + x_2^5 + x_3^{20} = 0$ at $p = 1601$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{20}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{20}\right)$	$172+1111\cdot 1601+690\cdot 1601^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^{17}$	p	p	1601
	$y_1^1 y_2^1 y_3^3 y_4^{13}$	p	p	1601
	$y_1^1 y_2^1 y_3^4 y_4^9$	p	p	1601
	$y_1^1 y_2^3 y_3^1 y_4^{11}$	p	p	1601
	$y_1^1 y_2^3 y_3^2 y_4^7$	p	p	1601
	$y_1^1 y_2^3 y_3^3 y_4^3$	p	p	1601
	$x_1^1 x_2^1 x_3^2 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{17}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{17}{20}\right)$	$222\cdot 1601+896\cdot 1601^2+\dots$
	$x_1^1 x_2^1 x_3^3 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{13}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{13}{20}\right)$	$559\cdot 1601+397\cdot 1601^2+\dots$
	$x_1^1 x_2^1 x_3^4 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{9}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{9}{20}\right)$	$1265\cdot 1601+806\cdot 1601^2+\dots$
	$x_1^1 x_2^2 y_3^1 y_4^{16} e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1601
	$x_1^1 x_2^2 y_3^2 y_4^{12} e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1601
	$x_1^1 x_2^2 y_3^3 y_4^8 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1601
	$x_1^1 x_2^2 y_3^4 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1601
	$x_1^1 x_2^3 x_3^1 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	1601
	$x_1^1 x_2^3 x_3^2 x_4^{12} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	1601
	$x_1^1 x_2^3 x_3^3 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	1601
	$x_1^1 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	1601
	$x_1^1 x_2^3 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{11}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{11}{20}\right)$	$81\cdot 1601+1157\cdot 1601^2+\dots$
	$x_1^1 x_2^3 x_3^2 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{7}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{7}{20}\right)$	$948\cdot 1601+572\cdot 1601^2+\dots$
	$x_1^1 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{20}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{20}\right)$	$613\cdot 1601+403\cdot 1601^2+\dots$
$H^{2,0}$	$x_1^1 x_2^3 x_3^4 x_4^{19} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{19}{20}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{19}{20}\right)$	$1480\cdot 1601^2+\dots$
$H^{2,2}$	$y_1^1 y_2^3 y_3^4 y_4^{19}$	p^2	p^2	2563201

Surface #5: $Q = (9, 6, 2, 1)$, $m = 18$

Equation: $x_0^2 + x_1^3 + x_2^9 + x_3^{18} = 0$

$p = 2917$

Table 5: Trace contributions for $x_0^2 + x_1^3 + x_2^9 + x_3^{18} = 0$ at $p = 2917$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{18}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{18}\right)$	$1304+854\cdot 2917+1837\cdot 2917^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^4 y_4^{13}$	p	p	2917
	$y_1^1 y_2^2 y_3^2 y_4^{11}$	p	p	2917
	$y_1^1 y_2^2 y_3^5 y_4^5$	p	p	2917
	$x_2^6 x_3^1 y_1^9 y_4^9 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	2917
	$x_2^2 x_3^1 y_1^9 y_4^9 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	2917
	$x_1^1 x_2^2 x_3^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{17}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{17}{18}\right)$	2917
	$x_1^1 x_2^3 x_3^{15} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	2917
	$x_1^1 x_2^3 x_3^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{13}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{13}{18}\right)$	$1131\cdot 2917+1533\cdot 2917^2+\dots$
	$x_1^1 x_2^3 x_3^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{11}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{11}{18}\right)$	2917
	$x_1^1 x_2^6 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	2917
	$x_1^1 x_2^7 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{7}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{7}{18}\right)$	$872\cdot 2917+1362\cdot 2917^2+\dots$
	$x_1^1 x_2^3 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)$	2917
	$x_1^1 x_2^3 x_3^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{13}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{13}{18}\right)$	2917
	$x_1^1 x_2^3 x_3^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{11}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{11}{18}\right)$	$2258\cdot 2917+2477\cdot 2917^2+\dots$
	$x_1^1 x_2^3 x_3^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)$	2917
	$x_1^1 x_2^3 x_3^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{7}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{7}{18}\right)$	2917
	$x_1^1 x_2^3 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{5}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{5}{18}\right)$	$717\cdot 2917+1068\cdot 2917^2+\dots$
	$x_1^1 x_2^3 x_3^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	2917
	$x_1^1 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{18}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{18}\right)$	2917
$H^{2,0}$	$x_1^1 x_2^8 x_3^{17} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{17}{18}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{17}{18}\right)$	$472\cdot 2917^2+\dots$
$H^{2,2}$	$y_1^1 y_2^2 y_3^8 y_4^{17}$	p^2	p^2	8508889

Surface #7: $Q = (6, 3, 2, 1)$, $m = 12$

Equation: $x_0^2 + x_1^4 + x_2^6 + x_3^{12} = 0$

$p = 577$

Table 7: Trace contributions for $x_0^2 + x_1^4 + x_2^6 + x_3^{12} = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{12}\right)$	$-\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{12}\right)$	$485+298\cdot 577+181\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^3 y_4^9$	p	p	577
	$y_1^1 y_2^1 y_3^5 y_4^5$	p	p	577
	$y_1^1 y_2^3 y_3^1 y_4^7$	p	p	577
	$y_1^1 y_2^3 y_3^3 y_4^3$	p	p	577
	$x_1^1 x_2^3 y_3^2 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^2 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^1 x_2^3 x_3^4 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	577
	$x_1^1 x_2^3 x_3^5 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{12}\right)$	$301\cdot 577+545\cdot 577^2+\dots$
	$x_1^1 x_2^3 y_3^2 y_4^8 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 y_3^4 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)$	577
	$x_1^1 x_2^3 x_3^2 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^1 x_2^3 x_3^5 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)$	577
	$x_1^1 x_2^3 x_3^7 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$554\cdot 577+495\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^5 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	577
$H^{2,0}$	$x_1^1 x_2^3 x_3^5 x_4^{11} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{11}{12}\right)$	$-p^2\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{11}{12}\right)$	$508\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^3 y_3^5 y_4^{11}$	p^2	p^2	332929

Surface #8: $Q = (4, 4, 3, 1)$, $m = 12$

Equation: $x_0^3 + x_1^3 + x_2^4 + x_3^{12} = 0$

$p = 433$

Table 8: Trace contributions for $x_0^3 + x_1^3 + x_2^4 + x_3^{12} = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$365+67\cdot 433+36\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^{10}$	p	p	433
	$y_1^1 y_2^1 y_3^3 y_4^7$	p	p	433
	$y_1^2 y_2^2 y_3^1 y_4^5$	p	p	433
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	433
	$x_1^1 x_2^1 x_3^2 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	433
	$x_1^1 x_2^1 x_3^3 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)$	$382\cdot 433+292\cdot 433^2+\dots$
	$x_1^1 x_2^2 y_3^1 y_4^9 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^1 x_2^2 y_3^2 y_4^6 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^1 x_2^2 y_3^3 y_4^3 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^1 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	433
	$x_1^1 x_2^2 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^1 x_2^2 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^2 x_2^1 y_3^1 y_4^9 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_2^1 y_3^2 y_4^6 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_2^1 y_3^3 y_4^3 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	433
	$x_1^2 x_2^2 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^2 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^2 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{12}\right)$	$416\cdot 433+224\cdot 433^2+\dots$
	$x_1^2 x_2^2 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	433
$H^{2,0}$	$x_1^2 x_2^2 x_3^3 x_4^{11} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{11}{12}\right)$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{11}{12}\right)$	$312\cdot 433^2+\dots$
$H^{2,2}$	$y_1^2 y_2^2 y_3^3 y_4^{11}$	p^2	p^2	187489

Surface #9: $Q = (4, 3, 3, 2)$, $m = 12$

Equation: $x_0^3 + x_1^4 + x_2^4 + x_3^6 = 0$

$p = 577$

Table 9: Trace contributions for $x_0^3 + x_1^4 + x_2^4 + x_3^6 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$485+298\cdot 577+181\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^4$	p	p	577
	$y_1^1 y_2^3 y_3^3 y_4^4$	p	p	577
	$y_1^2 y_2^1 y_3^1 y_4^5$	p	p	577
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	577
	$x_2^1 x_3^1 y_1^1 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_2^1 x_3^2 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_2^2 x_3^2 y_1^1 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_2^2 x_3^2 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_2^3 x_3^1 y_1^1 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_2^3 x_3^1 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_4^2 y_2^2 y_3^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$554\cdot 577+495\cdot 577^2+\dots$
	$x_1^2 x_4^2 y_2^2 y_3^2 e_1 e_4$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_2^2 x_1^1 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$301\cdot 577+545\cdot 577^2+\dots$
	$x_2^2 x_1^1 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_2^2 x_1^2 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^2 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
$H^{2,0}$	$x_1^2 x_2^3 x_3^3 x_4^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$508\cdot 577^2+\dots$
$H^{2,2}$	$y_1^2 y_2^3 y_3^3 y_4^5$	p^2	p^2	332929

Surface #10: $Q = (5, 2, 2, 1)$, $m = 10$

Equation: $x_0^2 + x_1^5 + x_2^5 + x_3^{10} = 0$

$$p = 3001$$

Table 10: Trace contributions for $x_0^2 + x_1^5 + x_2^5 + x_3^{10} = 0$ at $p = 3001$.

[illegible]

